328555(28)

B. E. (Fifth Semester) Examination, Nov.-Dec. 2019

(New Scheme)

(ET & T Engg. Branch)

ADVANCED MICROPROCESSOR and INTERFACING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) of each question is compulsory. Attempt any* ***two*** *part from (b), (c) and (d).*

Unit-I

1. (a) Explain the difference between 8086 and 8088 microprocessor. 2

(b) Explain the following PIN function of 8086 in detail. 7

(i) QSO and QS1

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(ii) RQ/GTO, RQ/GT1 (achive low)

(iii) Lock and test (active low)

(iv) S0, S1, S2

(c) What do you mean by pipelined architecture? How it is implemented in 8086 architecture? Explain it with register organization of 8086. 7

(d) Explain the architecture of 8089 IOP in detail. 7

Unit-II

2. (a) What is the functions of different types of REP prefix? 2

(b) The contents of different registers are given below. From physical addresses for different addressing modes, offset and 16 bit displacement = 5000 h, AX = 1000 H, DS = 2000H, Bx = 1000H, S1 = 3000H. 7

(c) Write a program to scans a string of words for an operand word specified in the register AX. The string is pointed by ES : DI : 2000H : 0500H? 7

(d) Explain the different types of interrupt in 8086 microprocessor. Explain with the help of interrupt vector table. 7

Unit-III

3. (a) Explain the difference between 8259IPC and 8259A IPC? 2

(b) Interface two 16x×8 EPROM and two 32 K×8 RAM chips with 8086. Select the starting address of EPROM suitably. The RAM address must start at 0000H. 7

(c) Design a programmable times using 8253 and 8086. Interface 8253 at an address 0040H for counter O and write an ALP for to generate a square wave of period 1 ms? The 8086 and 8253 run at 6 MHz and 1.5 MHz respectively? 7

(d) Explain the architecture of 8251 USART and also explain the control word register. 7

Unit-IV

4. (a) Explain the difference between 0808 and 0809 8 bit ADC. 2

(b) Interface ADC 0808 with 8086 using 8255 ports. USE port A of 8255 for transferring digital data output of ADC to the CPU and port C for control signals. Assume that an analog input is present at i/p 2 of the ADC and a clock input of suitable

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frequency is available for ACD. Draw the schematic and write required ALP? 7

(c) Explain the architecture of 8279 (Keyboard and Display Driver) in detail. 7

(d) Explain the interfacing of stepper motor with microprocessor 8086. 7

Unit-V

5. (a) How role of segment register becomes change when processor switched from real mode to protected mode? Define the format of selector. 2

(b) Explain the use of each of the following register of 80386 in detail. 7

(i) Control register

(ii) System address bus

(iii) Debug and test register

(c) What do you mean by paging? What are its advantages? Draw and discuss the paging mechanism of 80386 in detail. 7

(d) Explain the Architecture of 80386 in detail. 7

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